Assignment 2

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Task 1: Summary Table

Paper	Dataset	Image resolution	Aug. (Y/N)	Task type	DL/ML model	Novelty	Loss function	Eval metric	Eval score
0. Flood area segmentation using federated learning (Islam)	Kaggle Flood Area Segmentation, 290 drone images	512 × 512, grayscale	N	Segmentation	U-Net trained with FedAvg over 3 clients	Federated learning for flood mapping with non-IID clients, no raw data shared	BCE	Precision, Recall, F1, IoU, PA, MPA, FWIoU	Federated IoU 0.719, F1 0.836 (centralized IoU 0.745)
1. Multi-dataset + transfer learning for salt segmentation (Islam & Wali)	TGS, SEAM and F3 merged	TGS 101 × 101 (SEAM, F3 patch size not stated), all resized to 128 × 128 × 3	Y	Segmentation	EfficientNet-B7 (ImageNet) encoder with U-Net and LinkNet decoders	Training on 3 different seismic datasets together + pretrained encoder for better generalization	BCE + Dice	IoU, Precision, Recall, F1, Accuracy	Val IoU 0.955, F1 0.969. Test IoU: TGS 0.879, SEAM 0.932, F3 0.786 (U-Net)
2. TransFedSaltNet, federated transfer learning for salt (Islam & Wali)	TGS, SEAM, F3 (one dataset per client)	TGS 101 × 101, resized to 128 × 128 × 3	N	Segmentation	EfficientNet-B7 U-Net with FedAvg-FV (FedProx also compared)	Combines federated learning with transfer learning, plus federated validation for checkpointing	BCE + Dice (1:1)	IoU, Precision, Recall, F1, Accuracy	Held-out test IoU 0.874, F1 0.912 (centralized IoU 0.922)
3. Augmentation for brain tumor classification with limited data (Islam)	3 MRI datasets: Dataset1 (320, train), Dataset2 (394) and Dataset3 (3000) for testing	Resized to 256 × 256, channel copied to 3	Y	Classification (tumor / no tumor)	Pretrained ResNet50 (frozen) + dense head, DCGAN for synthetic images	Compares augmentation methods by testing on unseen, bigger datasets	Categorical cross-entropy	Accuracy, Precision, Recall, F1	Best on D2: geometric, 86.29% acc, F1 0.91. Best on D3: SMOTE, 84% acc, F1 0.86
4. Structure label prediction by retrieval and weak supervision (Alaudah et al.)	LANDMASS-2 (4000 images) and Netherlands F3 block	99 × 99	N	Image retrieval + weakly supervised pixel labelling	Curvelet-based similarity (authors' earlier work) + NMF with sparsity and orthogonality constraints (not DL)	Turns 1–2 exemplar images per class into thousands of pixel-level labels	NMF objective: $\|X - WH\|^2$ + orthogonality and L2 terms	Retrieval accuracy, MAP, AUC, Rand index; relative pixel accuracy for label mapping	Retrieval: RA 0.911, MAP 0.954, AUC 0.983, Rand 0.970. Label mapping: robustness curves and visual results, no single score
5. DL methods to classify salt bodies in seismic images (Islam)	TGS Salt Identification (Kaggle), about 4,000 images (from Fig. 4)	101 × 101, 1 channel	N	Segmentation	U-Net with SE-ResNet blocks, ELU activation	Adds SE (squeeze and excitation) blocks to ResNet U-Net, two stage loss training	BCE + Dice (300 epochs), then Lovász (300 epochs)	IoU (10-fold CV)	Avg val IoU 0.842 with Lovász (0.820 with BCE + Dice)
6. Automatic CV shortlisting tool (Mehboob et al.)	222 LinkedIn profiles (scraped) + 3 job descriptions	N/A (text)	N	Ranking / text matching	Bag of words + cosine similarity (no model training)	Web tool with HR and candidate portals that ranks CVs against a job post	None	Accuracy (match if similarity $\geq$ 70%)	99% claimed (14/14, 13/13, 49/49 matched)
7. Pix2Pix GAN for image-to-image translation (Henry et al.)	Maps dataset: New York satellite photos ↔ Google Maps, 1,097 train pairs	1200 × 600 pairs split to 256 × 256	N	Image translation (generation)	Pix2Pix cGAN: U-Net generator + 70 × 70 PatchGAN discriminator	No new method, it is a Keras re-implementation (tutorial) of Pix2Pix	Adversarial BCE + L1 (weighted 1:100)	None, only visual check	N/A (no numbers reported)
8. Federated Alternate Training, FAT (Mushtaq et al.)	KiTS19 (kidney CT), FeTS2021 (brain MRI)	Not stated (3D scans, nnU-Net default)	Y	Segmentation (federated, semi-sup.)	Pretrained nnU-Net, online / target (EMA) models on unlabelled silos	Alternates rounds between labelled and unlabelled silos, pseudo labels from target model	Dice + Cross-entropy	Dice score	KiTS19: kidney 0.951, tumor 0.730. FeTS: WT 0.913, TC 0.780, ET 0.660
9. Classical ML pipeline for forged receipts (Neerakan et al.)	Find it again! receipts (Kaggle), images + OCR text	256 × 256, grayscale	Y	Classification (forged / real)	LR, Random Forest, SVM, XGBoost on LBP, Canny edge and text features	Multimodal hand-crafted features (image + text) with SMOTE for heavy imbalance	Not stated (classical ML, SVM uses hinge loss)	Macro F1, Precision, Recall, PR-AUC, ROC-AUC, Accuracy	Tuned SVM: macro F1 0.543, PR-AUC 0.246, ROC-AUC 0.606

Task 2: Paper 5

Using deep learning based methods to classify salt bodies in seismic images, M. S. Islam, Journal of Applied Geophysics 178 (2020).

(a) Baseline paper

The paper never says its model beats any specific model. The “outperform” lines in the Abstract and Sec. 1 are about earlier machine learning and deep learning methods in general, and for its own result the paper only says a “promising” IoU was achieved. The closest thing to a baseline is Zeng et al. (2019). In Sec. 2.5 the paper says U-Net with ResNet blocks and IoU was used (Zeng et al., 2019; Wang et al., 2018), and then: “In this paper, we will modify this architecture using SE-ResNet.” Zeng’s model is the closest earlier model this work builds on, but its IoU is never shown next to 0.842. Table 1 lists Zeng on the SEAM dataset, and this paper uses TGS, so the numbers couldn’t be compared directly anyway.

(b) Information lost in preprocessing

As described, the pipeline never uses the depth value. Sec. 3.1 even says every image has a depth value that “turns out to be useful in training process”, but the loading step in Sec. 4.1 only reads the images and masks, and depth is never fed to the network as a channel or a second input, at least the paper does not mention it. Depth could help because the images come from “various locations chosen at random in the subsurface”, and the image itself cannot tell you where it came from.

(c) Most complicated operation to code

The Lovász loss would be the hardest part to set up. The paper only names it (Lovász-Softmax), cites Berman et al. (2018) and says “It is slower than other loss functions” (Sec. 4.8), but the detailed working of the loss is not given, so it has to be taken from the Berman paper. Using it also needs other changes: the last sigmoid has to be removed and the IoU threshold changed from 0.5 to 0 (Sec. 5). Since it is used for the second 300 epochs of all 10 folds on a single Colab K80 (Sec. 4.6), a slower loss adds a lot of training time.

(d) Trend that can’t be justified by the figure

The top plot (Fold 1, BCE + Dice, first 300 epochs) in Figure 7a. There are a couple of very steep declines in the blue line for the validation IoU. Around epoch 28 it falls to about 0.47, near 64 to about 0.54, near 97 to about 0.61, and near 127 again to about 0.47. The next time it bounces back the next epoch to about 0.73–0.78, the red training line remains smooth.

The text only states that the improvement with BCE becomes “negligible after certain time” and the caption simply states: K-Fold1 IOU score with BCE Vs Lovasz Loss. No reports of these crashes.

I would assume that’s more to do with the validation set than the model. Fig. 4 shows about 1,500 of the 4,000 images are in the empty or almost empty coverage class. If IoU is averaged over an image (paper doesn’t specify), some incorrect pixels Those can be the ones that take a score from 1 to 0 and drag the average down. However, the dips become less frequent after epoch 175, Perhaps it’s ReduceLROnPlateau that’s minimizing the learning rate, but that’s not mentioned either. And it’s not only fold 1. In fold Again, no comment, there is a dip near epoch 265 in 9 (Fig. 7b) where even the training IoU drops.